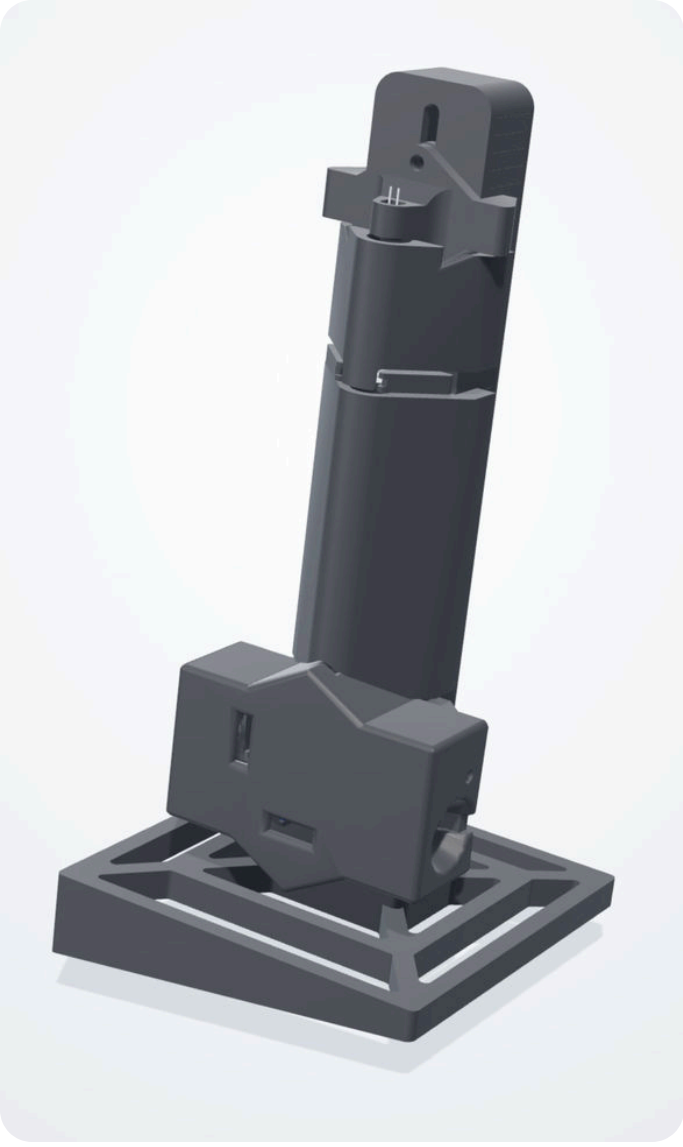
Pro Photometer

Assembly guide

An in-line OD600 meter for the bioreactor. A 600 nm beam is split 90 : 10, and two light sensors read the culture as it flows through a 2 mm cuvette, with no sample taken.



Questions or difficulties? Email **tau.noah@gmail.com**

READ FIRST

Before you start

SAFETY

- The soldering iron runs at 350 °C. Keep it in its stand and never touch the tip.
- Solder in a ventilated room or under a fume extractor. Do not breathe the smoke.
- Wear eye protection when you solder or cut wire.
- Unplug the Arduino before you change any wiring.
- Let the heat gun cool on a heat-safe surface.
- Heat-set inserts stay hot after you press them in. Let them cool before you touch them.
- The lens, the beam splitter and the cuvette are glass. Handle them gently.

Tools and supplies

- 3D printer (Bambu Lab X1 Carbon), Bambu Studio
- Soldering iron, wet sponge
- Heat gun
- Wire stripper and pliers
- Screwdriver for #10-32 screws
- Multimeter with a continuity (beep) setting
- Computer with Arduino App Lab
- Clean culture medium, for the blank
- Eye protection

Parts checklist

	QTY	PART	DETAILS
LIGHT PATH			
☐	1	**LED**	600 nm, 5 V, no internal resistors
☐	1	**LED clip**	black, holds the LED in the LED mount
☐	1	**Paralleling lens**	Plano-convex, SIGMAKOKI SLSQ-12.7-40P, ⌀12.7 mm, f = 40 mm
☐	1	**Beam splitter**	90 : 10 cube, about 12 × 12 × 14.5 mm
☐	1	**Flow cuvette**	2 mm inner, 4 mm outer diameter, 76 mm long
☐	2	**Lux sensors**	GY-302 (BH1750FVI): one sample, one reference
ELECTRONICS			
☐	1	**Arduino Uno Q**	
☐	1	**PCB**	2 × 8 cm prototype board
☐	1	**Transistor**	UTC PN2222
☐	3	**Resistors**	330 Ω, 1.2 kΩ and 10 kΩ, one of each
☐	12	**Wires, 20 AWG**	330 mm (12 + 1 in): 3 black, 3 red, 2 green, 2 blue. 610 mm (24 in): 1 red, 1 black
☐	2	**Heat-shrink tubes**	2.0 × 45 mm
☐	30	**2510 connectors**	Female: 2 × 5-pin, 4 × 2-pin. Male: 2 × 5-pin, 4 × 2-pin. 18 crimp pins
☐	20	**DuPont connections**	10 female crimp pins, 2 × 5-pin female housings
☐	10	**Male header pins, bent**	in one strip of 10
BODY			
☐		**Filament**	362.97 g black PLA and 43.77 g black TPU 95A
☐	7	**Screws, #10-32**	6 × 3/8 in (9.5 mm) and 1 × 1/2 in (12.7 mm)
☐	1	**Heat-set insert, #10-32**	brass, up to 3/8 in (9.5 mm) long, for a 5.6 mm hole

	QTY	PART	DETAILS
☐	1	**Roll of solder**	lead-free

Files

Print files	Download the STL files from the Design Files section of our wiki's Photometer page.
Inline Photometer app	v1.2.0, for Arduino App Lab on the Uno Q. Download it from the Code section of our wiki's Photometer page.

STAGE 1 Print the parts

YOU NEED

- 3D printer with black PLA and black TPU 95A
- The 8 STL files

Printer	Bambu Lab X1 Carbon, 0.4 mm nozzle, textured PEI plate	**Profile**	0.20 mm Standard @BBL X1C: layer height 0.2 mm, first layer 0.2 mm
Top surface	Line width 0.25 mm	**Infill**	20 %, gyroid
Supports	Normal (auto), 30° threshold	**Filament**	Black PLA for every part. TPU 95A painted on the seats of the lens holder and the sensor frame

1. Download the 8 STL files from the Design Files section of our wiki's Photometer page.
2. Open them in Bambu Studio, with the settings in the table.
3. Put the lens holder ❷ and the sensor frame ❹ on plate 1.
4. Set plate 1's filament to black PLA.
5. Paint TPU 95A onto the seats of these two parts.
   The seats are where the lens, beam splitter, cuvette and sensors sit; the TPU grips them.
6. Put the other six parts on the other plates, in black PLA.
7. Slice each plate and print it.
8. Remove the supports.



❶ LED mount



❷ Lens holder



❸ Body



❹ Sensor frame



❺ Base



❻ Body cover



❼ Body cover, small



❽ Sensor frame cover

CHECK

1. You have all 8 printed parts. The lens holder and the sensor frame have TPU on their seats.

STAGE 2 Make the LED cable

YOU NEED

- The 600 nm LED; the red and black 610 mm (24 in) wires; 2 heat-shrink tubes; a 2-pin 2510 female connector and 2 crimp pins
- Wire stripper, pliers, soldering iron, heat gun

1. Twist the red and black 610 mm wires together.
   Leave about 75 mm (3 in) untwisted at each end.
2. At one end, strip 25 mm (1 in) off each wire.
3. Slide a heat-shrink tube onto each wire, well back from the bare end.
4. Wrap the bare red wire around the LED's long leg (+).
5. Wrap the bare black wire around the LED's short leg (−).
6. Heat the soldering iron to 350 °C.
7. Solder each joint: hold the iron on the joint to heat it, then feed solder into the joint until it flows round both wires.
   Feed the solder to the joint, not to the iron. Take the solder away first, then the iron.
8. Let the joints cool for 30 seconds.
9. Slide each heat-shrink tube over its bare joint.
10. Shrink both tubes with the heat gun.
11. Twist the wires tight again.
12. At the other end, strip 8 mm (0.3 in) off each wire.
13. Lay the red wire in a 2510 crimp pin: large tabs on the insulation, small tabs on the bare wire.
14. Fold the large tabs with pliers so they grip the insulation.
15. Fold the small tabs with pliers so they press onto the bare wire.
16. Solder the small tabs to the wire.
    Keep solder off the large tabs, or the pin will not seat.
17. Crimp the black wire the same way (steps 13 to 16).
18. Push the red pin into the left slot of a 2-pin 2510 female connector until it clicks. Push the black pin into the right slot.
    Tug each wire: it must not pull out.

CHECK

1. Push the LED connector onto a 2-pin 2510 male connector. It must seat fully.

If it fails: Solder ran onto the large tabs of a crimp pin and blocks it. Redo that pin.

STAGE 3 Make the two sensor cables

YOU NEED

- 10 wires, 330 mm (12 + 1 in): 3 black, 3 red, 2 green, 2 blue
- 10 DuPont female crimp pins and 2 × 5-pin female housings
- 10 × 2510 crimp pins and 2 × 5-pin 2510 female connectors
- For the checks: the strip of 10 bent header pins and 2 × 5-pin 2510 male connectors
- Wire stripper, pliers, soldering iron

1. Cut 25 mm (1 in) off each of the 10 wires.
   Keep the 10 short pieces: they are the PCB's jumpers (Stage 4).
2. Strip 8 mm (0.3 in) off both ends of the 10 long wires.
3. Crimp a DuPont female pin onto one end of each wire, as in Stage 2.
   Large tabs on the insulation, small tabs on the wire, then solder the small tabs.
4. Push the pins into the two DuPont housings until they click, in the order below.

	ADDR	SDA	SCL	GND	VCC (3.3 V)
Housing 1 · sample ADDR to 3.3 V · I²C 0x5C	red	blue	green	black	red
Housing 2 · reference ADDR to GND · I²C 0x23	black	blue	green	black	red

> Read each housing from its ADDR end. This is the GY-302's own pin order, VCC, GND, SCL, SDA, ADDR, reversed: check it against the labels printed on your board before you plug a housing in. The red VCC wires go to the Uno Q's 3.3 V; SDA goes to D20 and SCL to D21.

CHECK · TEST 1

1. Push each DuPont housing onto the bent header pins. It must fit.
2. Pull the header pins out again.

If it fails: Solder on a crimp pin ran too close to the insulation and blocks the header pin. Redo the pins of that housing.

5. Crimp a 2510 pin onto the other end of each wire, the same way.
6. Push the pins into the two 5-pin 2510 female connectors until they click.
   Keep each cable's wires in the same order as its DuPont housing. Never mix the two cables in one connector.
7. Label the cables: sample (housing 1) and reference (housing 2).

CHECK · TEST 2

1. Push each 2510 female connector into a 5-pin 2510 male connector. They must click together.
2. Take the male connector off again.

If it fails: Solder on a crimp pin ran too close to the insulation and blocks the connector. Redo that connector's pins.

STAGE 4 Solder the sensors and the PCB

YOU NEED

- Both GY-302 sensors
- The strip of 10 bent header pins
- The 2 × 8 cm PCB
- The six 2510 male connectors
- Resistors: 330 Ω, 1.2 kΩ and 10 kΩ
- The PN2222 transistor
- The 10 short wires from Stage 3
- Soldering iron, solder, wire stripper, multimeter

THE SENSORS

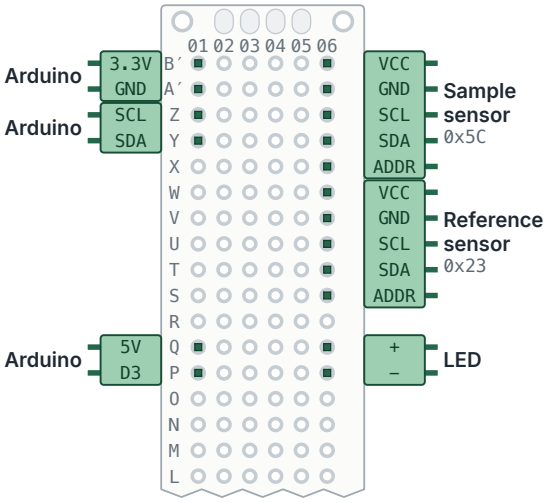
1. Snap the header strip into two strips of 5 pins.
2. Put one strip into the five holes of a GY-302.
3. Solder each pin to its pad: heat the pin and the pad together, then feed solder into the joint.
   Keep the iron on the board briefly: too long melts the plastic and spoils the joint.
4. Check that no solder joins two pins.
5. Solder the second sensor the same way.

CHECK · THE SENSORS

1. On both sensors, every pin is soldered and no solder joins two pins.

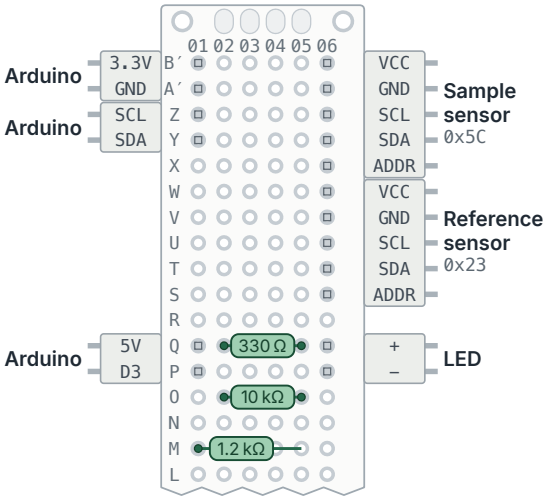
THE PCB

Hold the PCB as in the pictures, columns 01 to 06 from left to right. Its rows are lettered up from the bottom edge, A to Z, then A and B again: this guide calls those top two **A′** and **B′**. A hole is named by row, then column: Q02 is row Q, column 02. Parts go in from the component side and are soldered on the solder side, underneath. Green in each picture is what the step adds.



Component side. Green: the six connectors.

6. Push the six 2510 male connectors into the PCB from the component side, their pins pointing out past the edge.
   Left edge, column 01: three 2-pin connectors for the Arduino, in rows B′–A′, Z–Y and Q–P. Right edge, column 06: the sample sensor's 5-pin connector in rows B′ to X, the reference sensor's in rows W to S, and the LED's 2-pin connector in rows Q–P.
7. Turn the board over. Solder one pin of each connector, check that the connector sits flat, then solder its other pins.
   18 pins in all.

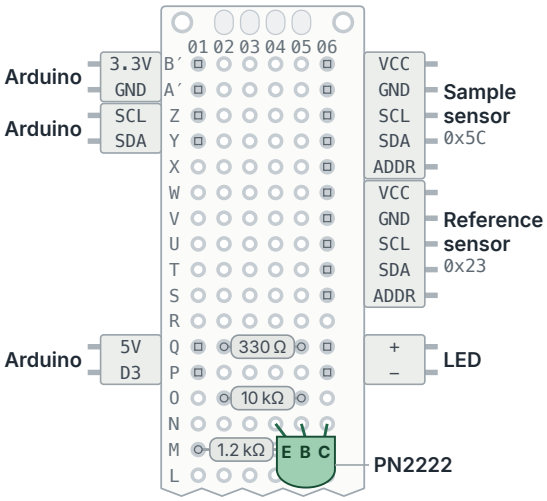


Component side. Green: the three resistors.

8. Bend each resistor's leads down and push them through from the component side, into the holes below.
   The 1.2 kΩ sits to the left; its right lead runs on to M05.

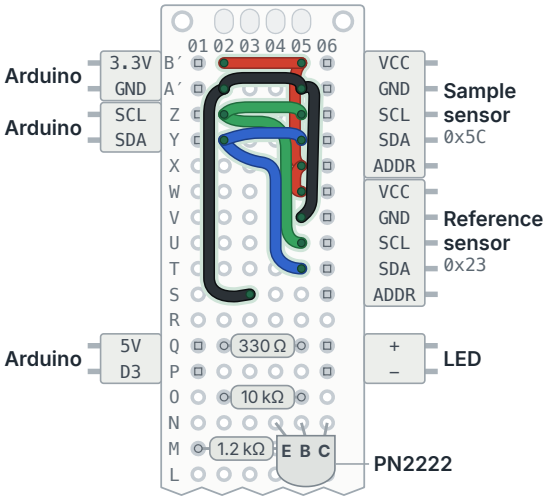
RESISTOR	BANDS	HOLES
330 Ω	orange, orange, brown, gold	Q02, Q05
10 kΩ	brown, black, orange, gold	O02, O05
1.2 kΩ	brown, red, red, gold	M01, M05

9. Solder each lead on the solder side, then trim it.



Component side. Green: the transistor.

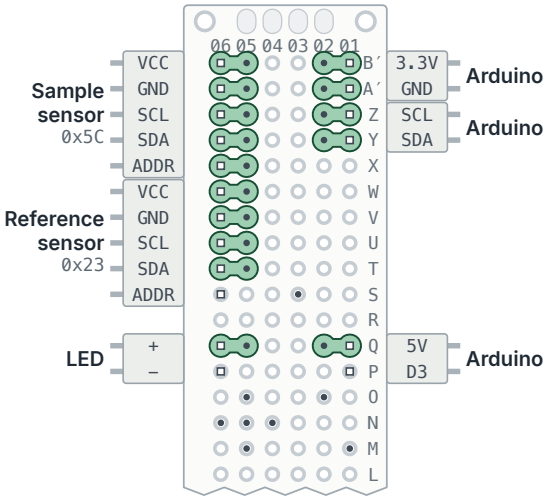
10. Find the emitter (E), base (B) and collector (C) of your PN2222 in its datasheet.
    Check before you solder: the pin order differs between makes, E-B-C on some and C-B-E on others.
11. Push E into N04, B into N05 and C into N06, from the component side. Fold the body down flat over rows M and L.
    E goes to ground, B to the two resistors, C to the LED's – pin.
12. Solder the three legs and trim them.



Component side. The ten jumpers, in their colours.

13. Cut the 10 short wires from Stage 3 to fit the routes in the picture. Strip 3 mm (0.1 in) off each end.
14. Push both ends of each wire through from the component side, as in the table, and solder them.
    Six holes take two wire ends: B′05, X05, A′02, A′05, Z02 and Y02. Put both ends in before you solder; if the second will not fit, solder it onto the first one's joint on the component side.

COLOUR	CARRIES	EACH WIRE, FROM → TO
red	3.3 V	B′02 → B′05, B′05 → X05, X05 → W05
black	GND	A′02 → A′05, A′05 → V05, A′02 → S03
green	SCL	Z02 → Z05, Z02 → U05
blue	SDA	Y02 → Y05, Y02 → T05



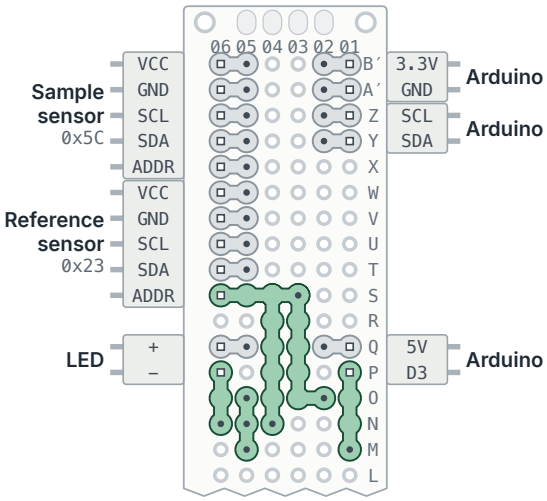
Solder side, turned over. Green: the 15 bridges.

15 Turn the board over, left to right, so that row B′ stays at the top.

Column 06 is now on the left, as in the picture.

16 Bridge each connector pin to the hole beside it: heat both pads together and feed in solder until it joins them.

15 bridges: column 06 to 05 in rows B′ to T, column 01 to 02 in rows B′ to Y, then Q06 to Q05 and Q01 to Q02.



Solder side. Green: the six traces.

17 Make the six traces: run solder from hole to hole along each green line.

On the long ones, a trimmed lead laid along the holes and soldered at each makes it easier.

TRACE	HOLES	JOINS
Row S	S06 to S03	the reference ADDR to GND
Column 04	S04 to N04	GND to the transistor's E
Column 03	S03 to O03, O02	GND to the 10 kΩ
Column 06	P06 to N06	the LED's – to C
Column 05	O05 to M05	B to both resistors
Column 01	P01 to M01	D3 to the 1.2 kΩ

18 Check that no solder joins two lines the picture keeps apart.

Look hardest where they run side by side: columns 04 to 06 in rows Q to N, and rows S and T.

CHECK · THE PCB

Set a multimeter to continuity (beep) and test on the connectors' pins. The three connectors on the left edge are the Arduino's.

1. **GND:** the left GND pin beeps to both sensors' GND pins and to the reference sensor's ADDR pin.
2. **3.3 V:** the left 3.3 V pin beeps to both sensors' VCC pins and to the sample sensor's ADDR pin.
3. **SDA, SCL:** each beeps from the left edge to the same pin on both sensor connectors.
4. **No shorts:** no beep between any two of GND, 3.3 V, 5 V, SDA and SCL; between 5 V and D3; or between the LED's + pin and GND or its – pin.
5. **330 Ω:** on the Ω setting, the left 5 V pin reads about 330 Ω to the LED's + pin.

If it fails: A missing beep is a missing bridge or a poor joint: reheat it and add a little solder. An extra beep is solder joining two holes that should stay apart: reheat it and draw the extra solder away with the iron's tip.

STAGE 5 Connect the sensors and test them

YOU NEED

- Both sensors, with their header pins
- The two sensor cables and the LED cable
- The PCB and the Arduino Uno Q
- A computer with Arduino App Lab
- The Inline Photometer app, v1.2.0

1. Push housing 1 onto the sample sensor's pins and housing 2 onto the reference sensor's pins.
2. Plug the cables into the PCB's 2510 male connectors on its right edge: the sample cable (housing 1) into the top 5-pin connector, the reference cable (housing 2) into the one below it, and the LED cable into the 2-pin connector.
   The PCB sets each sensor's address (ADDR to 3.3 V on the top connector, 0x5C; to GND on the other, 0x23), so each cable must go to its own connector. Each sensor cable reads red, black, green, blue from the top pin down; the LED cable's red wire is on the top pin.
3. Connect the three 2-pin connectors on the PCB's left edge to the Arduino Uno Q, as in the table below.

PCB PIN	UNO Q PIN	CARRIES
3.3 V	3.3 V	the sensors' supply (the red wires)
GND	GND	ground
SCL	D21	the I²C clock (the green wires)
SDA	D20	the I²C data (the blue wires)
5 V	5 V	the LED's supply
D3	D3	the LED switch: to the transistor's base, through 1.2 kΩ

4. In Arduino App Lab, make a new app. Copy in **app.yaml**, the **sketch** and **python** folders, and **assets/index.html**.
   The python folder must include photometer_algos.py.
5. Add the brick **WebUI – HTML**.
6. Add the sketch library **Arduino_RouterBridge**.
7. Click **Run** and read the [boot] lines in the console.

```
[boot] step 2/4: I2C probe 0x5C / 0x23
[boot]   sample    0x5C: ok (0 ms)
[boot]   reference 0x23: ok (0 ms)
```

CHECK · TEST 3

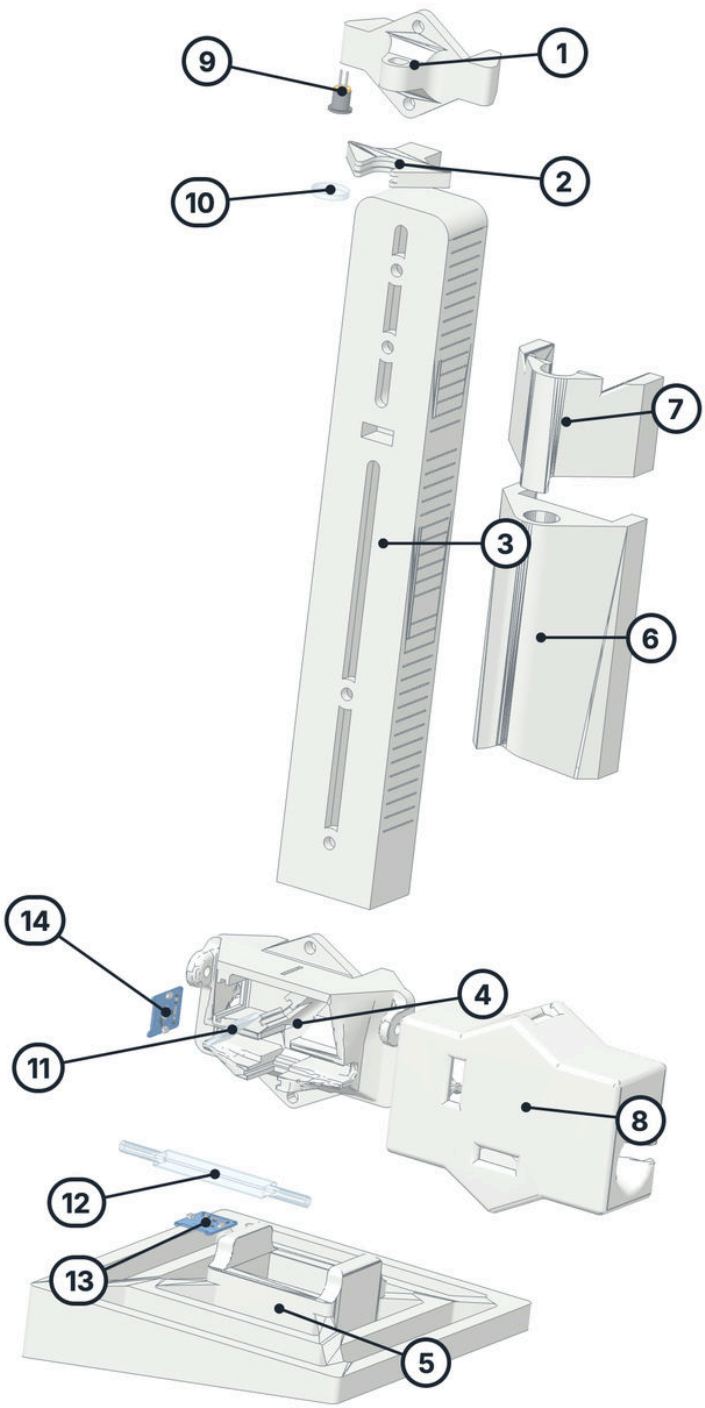
1. The console shows **sample 0x5C: ok** and **reference 0x23: ok**.
2. In a bright room, cover each sensor with a finger in turn.
3. Its reading drops from a high value to near zero.

If it fails: nak means no device at that address: no power, or ADDR wired wrong (check the housing order). **timeout** means a bus problem: SDA or SCL shorted or pulled low, or GND missing. A failed sensor is retried every 10 seconds, so it comes back once the wiring is fixed.

STAGE 6 Assemble the body

YOU NEED

- The 8 printed parts
- The LED on its cable, and its clip
- The lens, the beam splitter and the flow cuvette
- Both sensors, on their cables
- 7 × #10-32 screws: 6 × 3/8 in, 1 × 1/2 in
- The #10-32 heat-set insert, and the soldering iron to press it in
- Screwdriver



The parts, numbered as on the wiki (Photometer page, 6.2). They go together in the reverse order of the exploded view: the optics into their holders, the column onto its base, then the covers.

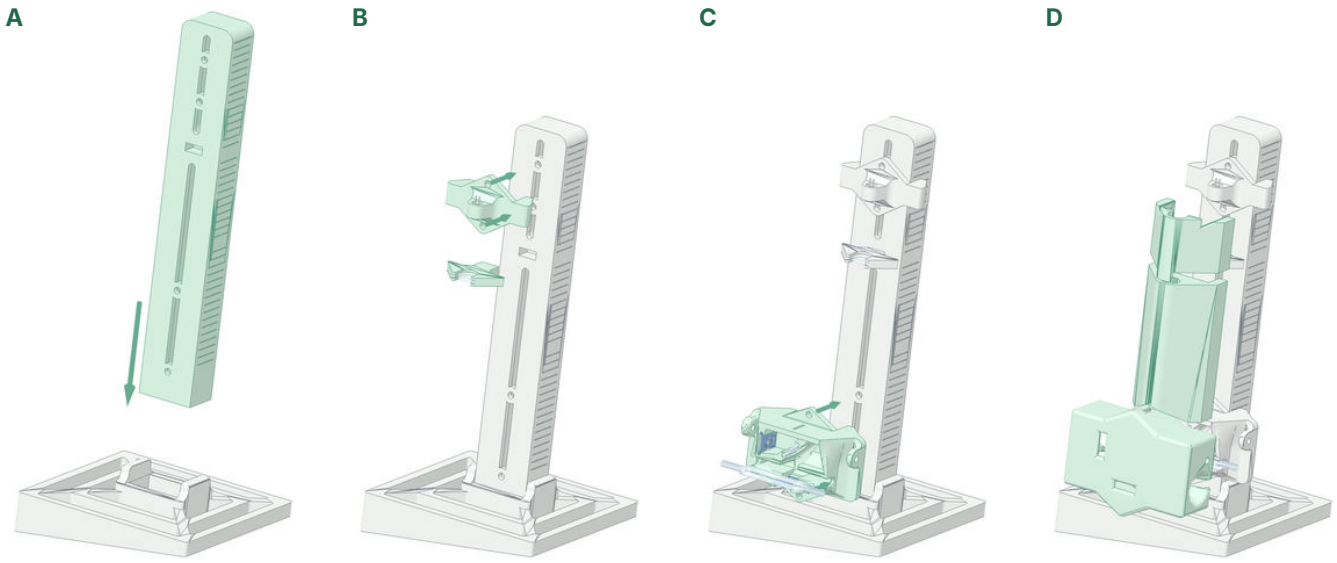
PRINTED

1. LED mount
2. Lens holder
3. Body
4. Sensor frame
5. Base
6. Body cover
7. Body cover, small
8. Sensor frame cover

BOUGHT

9. 600 nm LED, in its clip
10. Paralleling lens
11. Beam splitter, 90 : 10
12. Flow cuvette
13. Sample sensor (GY-302)
14. Reference sensor (GY-302)

Green in pictures A to D marks the parts each group of steps adds.



A the body into the base (steps 1–3). **B** the LED mount and the lens holder (steps 4–8). **C** the sensor frame with its optics (steps 9–14). **D** the three covers (steps 15–19).

THE COLUMN · A, B

1. Press the heat-set insert into the pocket in the foot of the body ❸, with the hot soldering iron, until it sits flush.
   Let it cool.
2. Push the bottom of the body into the socket on the base ❺, all the way down.
3. Turn the two over. Drive the 1/2 in screw up through the hole in the base's floor into the insert.
4. Press the LED ⑨, in its clip, into the round hole of the LED mount ❶.
   The LED points down the column; its legs and cable come out of the top.
5. Hold the LED mount on the top of the body, its back plate flat on the body's front.
6. Fix it with 2 × 3/8 in screws, through the two holes in its back plate into the body.
7. Lay the lens ⑩ in the round seat of the lens holder ❷.
8. Push the lens holder's tab into the slot in the body's front, below the LED mount.

THE OPTICS · C

9. Put the beam splitter ⑪ in the square seat of the sensor frame ❹.
   Turn it so it reflects light toward the side sensor slot.
10. Press the reference sensor ⑭ into the side slot, its chip facing the beam splitter.
11. Press the sample sensor ⑬ into the slot below, its chip facing up.
12. Hold the sensor frame on the body, just above the base, its back plate flat on the body's front.
13. Fix it with 2 × 3/8 in screws, through the two holes in its back plate into the body.
14. Lay the flow cuvette ⑫ in its cradle across the sensor frame, between the beam splitter and the sample sensor.
    It is glass and breaks easily: handle it gently.

15. Snap the small body cover (7) onto the body over the lens holder, from the front.
16. Snap the body cover (6) onto the body below it, from the front.
17. Fit the sensor frame cover (8) over the sensor frame, its side ears outside the frame's ears.
18. Fix it with 2 × 3/8 in screws, one through each side ear into the frame's ear.
19. Lead the sensor cables out through the cover's two windows.

STAGE 7 Align and blank

YOU NEED

- The assembled photometer, wired up and running the app
- Clean culture medium

1. In the app's light-path mode, choose **beam splitter** and enter the split ratio **9 : 1**.
2. Turn on alignment mode. The LED stays on.
3. Turn the LED a little in its clip until the sample-to-reference ratio is about 9 : 1.
   Both readings should be high, but not at 100 %.
4. Fill the flow cuvette with clean medium.
5. Press **Calibrate blank**.
   In the bioreactor, blank only after sterilizing and rinsing (bioreactor guide, Stage 6).

CHECK · TEST 4

1. The sample-to-reference ratio holds at about 9 : 1.
2. Neither reading is at 100 %.

If it fails: Turn the LED in its clip again and recheck. The app also warns when the ratio is outside half to double the ratio you entered. Blank again whenever the light path has been moved.